

$$\textcircled{1} a) \nabla f(x, y) = (-2x + y - 2, x + 4y + 1)$$

$$-2x + y - 2 = 3\lambda$$

$$x + 4y + 1 = -6\lambda$$

$$\underline{-1 = -\lambda \quad \dots \lambda = 1}$$

$$\left. \begin{array}{l} -2x + y - 2 = 3 \\ x + 4y + 1 = -6 \end{array} \right\} 9y = -9 \quad \dots y = -1$$

$$x = -9 - 4(-1) = -3$$

Probleme $f(-3, -1) = -9 + 3 + 2 + 6 - 1 = 1$, je

$$a = (-3, -1, 1).$$

$$b) \nabla f(-2, 1) = (3, 3) \Rightarrow$$

$$\Rightarrow h = - \frac{\nabla f(-2, 1)}{\|\nabla f(-2, 1)\|} = - \frac{1}{\sqrt{2}} (1, 1)$$

② Hledáme extrém body exobem funkce
 $f(x, y) = (x - \frac{1}{2})^2 + y^2$ na M .

$$2(x - \frac{1}{2}) + \lambda \frac{1}{2}x = 0$$

$$2y + \lambda y = 0 \dots y(2 + \lambda) = 0 \begin{cases} y = 0 \\ \lambda = -2 \end{cases}$$

$$\frac{x^2}{4} + \frac{y^2}{2} = 1$$

• $y = 0 \Rightarrow x^2 = 4 \dots x = \pm 2$

$$f(2, 0) = \frac{9}{4}, \quad f(-2, 0) = \frac{25}{4}$$

• $\lambda = -2 \Rightarrow 2x - 1 - x = 0$
 $x = 1$

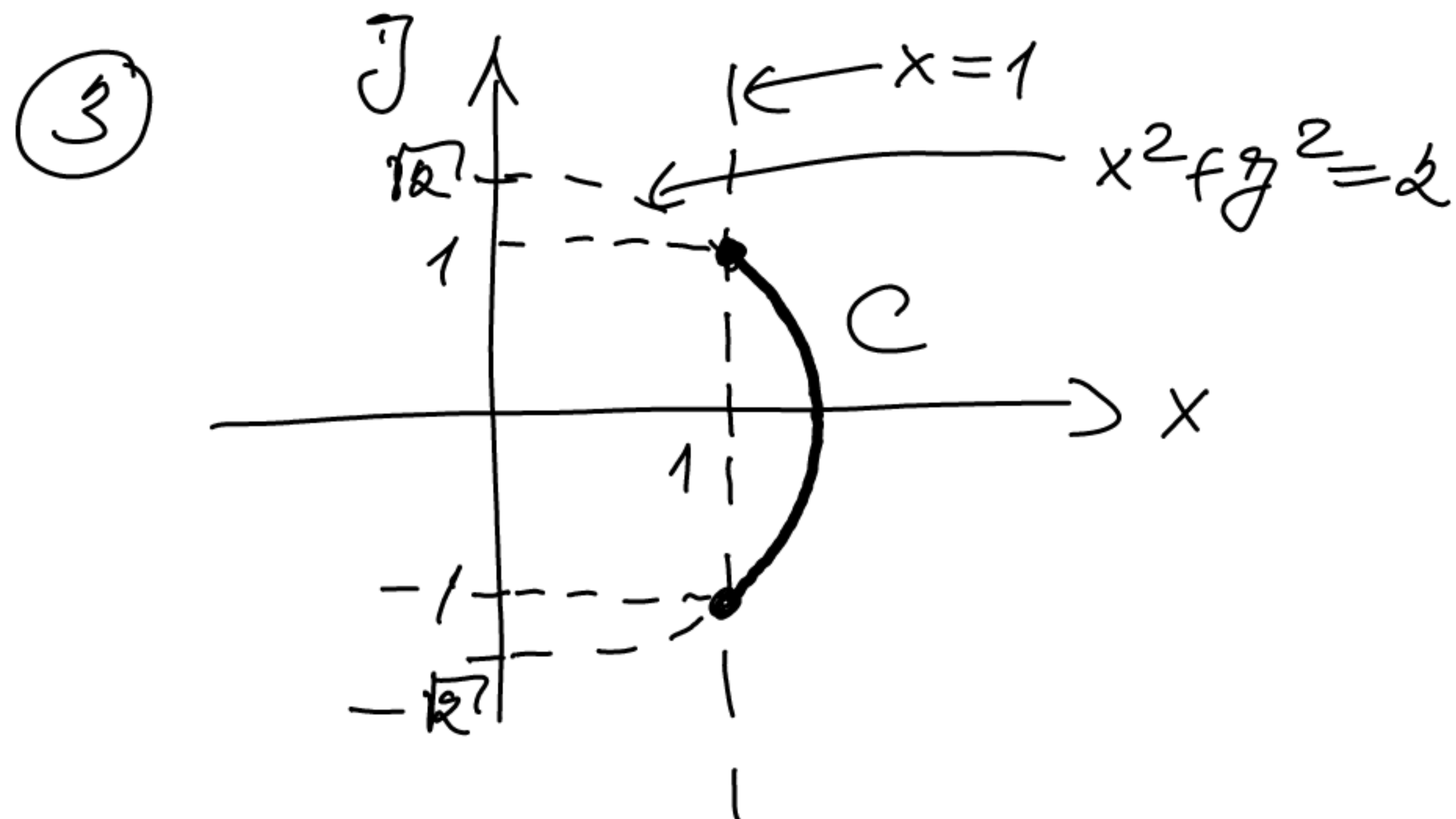
$$\frac{y^2}{2} = 1 - \frac{1}{4} = \frac{3}{4}, \text{ tedy } y = \pm \sqrt{\frac{3}{2}}$$

$$f(1, \pm \sqrt{\frac{3}{2}}) = \frac{1}{4} + \frac{3}{2} = \frac{7}{4}$$

Porovnáním funkčních hodnot
v požadovaných bodech vidíme, že

a) nejblíže od bodu $(\frac{1}{2}, 0)$ jsou body
 $(1, \sqrt{\frac{3}{2}})$ a $(1, -\sqrt{\frac{3}{2}})$;

b) nejdalek od bodu $(\frac{1}{2}, 0)$ je bod $(-2, 0)$.



a) $\varphi(t) = (\sqrt{2} \cos t, \sqrt{2} \sin t)$, $t \in [-\frac{\pi}{4}, \frac{\pi}{4}]$

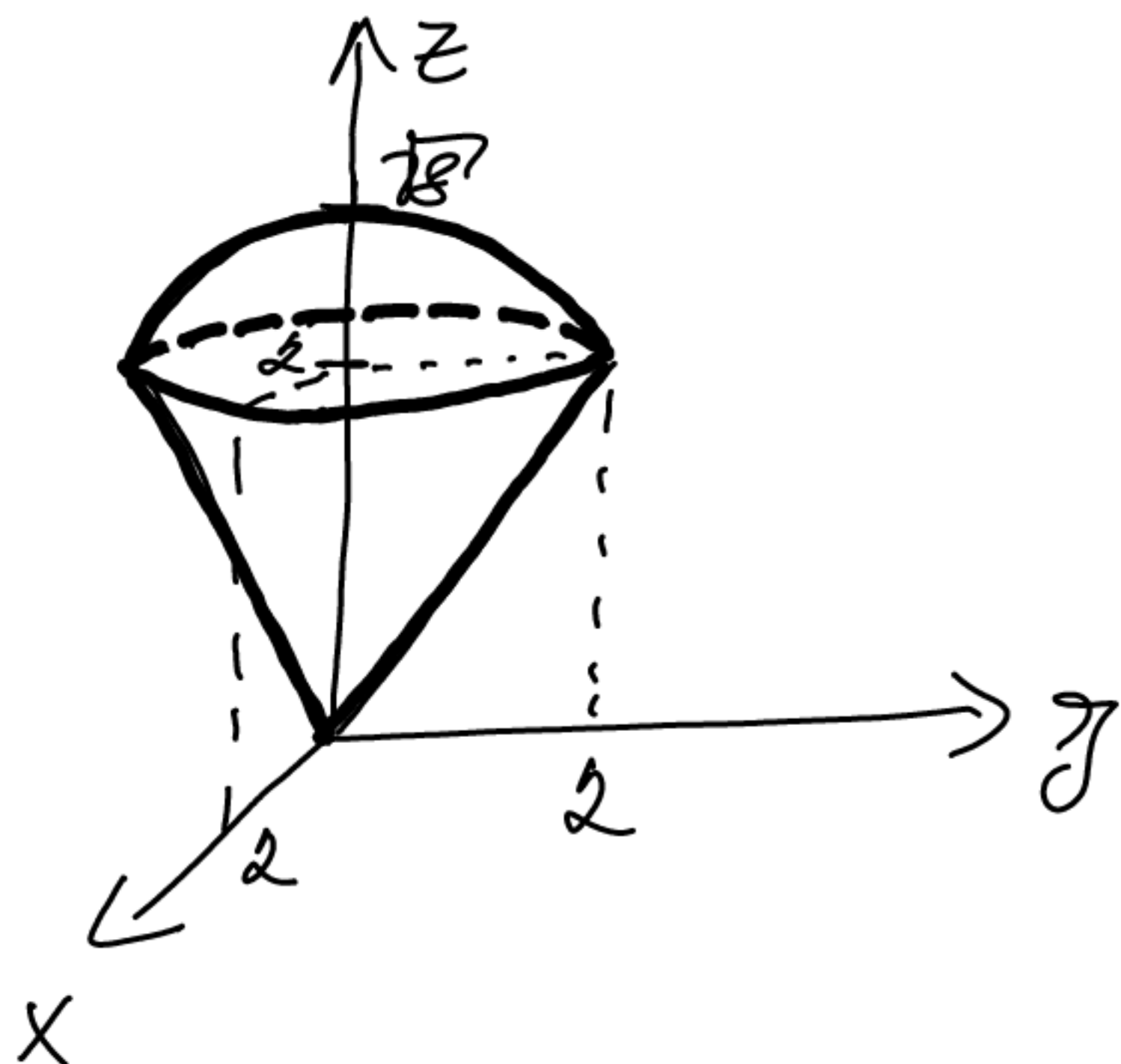
b) $\|\varphi'(t)\| = \|(-\sqrt{2} \sin t, \sqrt{2} \cos t)\| = \sqrt{2}$

$\Rightarrow \int_C x y^2 ds = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{2} \cos t \cdot 2 \sin^2 t \cdot \sqrt{2} dt$

$= 4 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos t \sin^2 t dt = 8 \int_0^{\frac{\pi}{4}} \cos t \sin^2 t dt$

$= 8 \left[\frac{\sin^3 t}{3} \right]_0^{\frac{\pi}{4}} = \frac{8}{3} \left(\frac{\sqrt{2}}{2} \right)^3 = \frac{2\sqrt{2}}{3}$

④



$$\nabla \cdot F(x, y, z) = -z + 0 + 2z = z$$

$$\int_S F(x, y, z) \cdot d\vec{\sigma} = \int_V \nabla \cdot F(x, y, z) dV$$

$$= \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^2 r \cos \theta \, r^2 \sin \theta \, dr \, d\theta \, d\varphi$$

$$= \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \left[\frac{r^4}{4} \right]_0^2 \sin \theta \cos \theta \, d\theta \, d\varphi$$

$$= 16 \int_0^{2\pi} \left[\frac{\sin^2 \theta}{2} \right]_0^{\frac{\pi}{4}} d\varphi = 16\pi \left(\frac{1}{2} \right)^2$$

$$= 8\pi$$

$$\textcircled{5} \lim_{\ell \rightarrow \infty} \sqrt[\ell]{\frac{1}{3^\ell \ell} |x+2|^\ell} = \lim_{\ell \rightarrow \infty} \frac{1}{3 \sqrt[\ell]{\ell}} |x+2|$$

$$= \frac{|x+2|}{3} < 1 \Leftrightarrow |x+2| < 3$$

Polarna konvergenca je $R=3$.

Ostalo vidimo, če nima nekonečnosti
v 2 (interval konvergenca je $(-5, 1)$).

$$\sum_{\ell=1}^{\infty} \frac{1}{3^\ell \ell} (x+2)^\ell \sim \sum_{\ell=1}^{\infty} \frac{1}{3^\ell} (x+2)^{\ell-1}$$

$$= \sum_{\ell=0}^{\infty} \frac{1}{3^{\ell+1}} (x+2)^\ell = \frac{1}{3} \sum_{\ell=0}^{\infty} \left(\frac{x+2}{3} \right)^\ell$$

$$= \frac{1}{3} \frac{1}{1 - \frac{x+2}{3}} = \frac{1}{1-x}$$

Na intervalu konvergenca naj bo:

$$\sum_{\ell=1}^{\infty} \frac{1}{3^\ell \ell} (x+2)^\ell = \int \frac{1}{1-x} dx = -\ln(1-x) + C$$

$$x = -2 \dots 0 = -\ln 3 + C \Leftrightarrow C = \ln 3$$

$$\text{Tedy } \sum_{\ell=1}^{\infty} \frac{1}{3^\ell \ell} (x+2)^\ell = -\ln(1-x) + \ln 3$$

(pro $|x+2| < 3$)